

## Dual Helmholtz resonator

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### ABSTRACT

Acoustic characteristics of a dual Helmholtz resonator which consists of a pair of cylindrical neck and cavity connected in series (neck–cavity–neck–cavity) are established primarily in terms of a lumped-parameter theory. The results are then compared to (a) a two-dimensional analytical approach by applying a pressure/velocity matching technique for a piston-driven model; (b) a three-dimensional boundary element method for a pipe-mounted model; and (c) experiments on an impedance-tube test setup with one fabricated prototype installed in a sidebranch orientation (pipe-mounted model). Closed-form expressions have been developed in the present study for the resonance frequencies and the transmission loss of this two degree-of-freedom system based on the Newton's second law of motion for a lumped system. The resonance frequencies and the transmission loss from this theory agree well with the analytical results, while showing a reasonable comparison with both the numerical predictions and the measurements.

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### 1. Introduction

Helmholtz resonators, which consist of a cavity communicating with the main duct through a neck, have been widely used to effectively attenuate the narrow-band low-frequency noise. The classic lumped approach approximates this resonator as an equivalent spring (cavity) and mass (neck) system, and yields the expressions for the resonator frequency and the transmission loss [1–5]. A number of studies [6–10] have also developed a one-dimensional analytical approach to investigate the wave attenuation properties. To account for the nonplanar wave propagation in both the neck and the cavity, multi-dimensional analytical approaches have also been employed to predict the sound attenuation in Helmholtz resonators with circular concentric cavity [10], circular asymmetric cavity [11], extended neck [12], and lined with absorbing material [13].

To improve the sound attenuation performance, Helmholtz resonators with a variety of modifications have been examined. Selamet and Lee [12] studied the effect of length, shape, and perforation of the neck extension on the resonance frequency and transmission loss of concentric circular Helmholtz resonators with extended neck. Selamet et al. [13] developed a closed-form, 2D analytical solution to investigate the effect of density and thickness of the fibrous material in the cavity on the resonance fre-

quency and transmission loss of circular Helmholtz resonators lined with absorbent. Tang [14] experimentally and theoretically investigated the Helmholtz resonators with tapered necks with the cross-sectional area increasing towards cavity. Griffin et al. [15] developed an analytical model for a single, coupled resonator system mounted on a one-dimensional duct. The proposed mechanically-coupled resonators produced a particular transmission loss response, provided a wider bandwidth of attenuation, and adapted the transmission loss characteristics of a structure to attenuate disturbances of varying frequency. Wan and Soedel [16] derived an expression for the resonance frequencies of 2-DOF Helmholtz resonator using a lumped analysis. They obtained the resonance frequencies and compared them to the computational results and measurements. De Bedout et al. [17] proposed a tunable Helmholtz resonator with a feedback system to achieve an optimal tuning of time-varying tonal noise.

Helmholtz resonators formed by a pair of neck and cavity connected in series (neck–cavity–neck–cavity) lead to two resonances at low frequencies. The two low frequencies they can control combined by the space constraints make dual Helmholtz resonators desirable silencing elements, for example, in engine breathing systems. Yet, the acoustic attenuation performance of such noise attenuators remains to be further established. The present work therefore concentrates on the acoustic characteristics (including both the resonance frequencies and the transmission loss) of dual Helmholtz resonators. The objective is to investigate the acoustic attenuation performance of dual Helmholtz resonators by: (1) developing analytical expressions for both the resonance frequencies and the transmission loss for a piston-driven model in terms of the lumped-parameter theory; (2) devising a two-dimensional

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analytical method for the piston-driven model using a pressure/velocity matching technique; (3) performing computations by implementing the Boundary Element Method (BEM); and (4) comparing the lumped and two-dimensional results with the numerical predictions and measurements with a fabricated base model. In approaches (1)–(3), the effect of viscous dissipation through the necks will be ignored for simplicity. Such an assumption may be anticipated to influence comparisons with experiments in the immediate vicinity of resonances.

Following this Introduction, Sections 2 and 3 respectively develop the lumped-parameter theory and the two-dimensional analytical method for the piston-driven model; Section 4 describes the BEM; Section 5 compares the lumped and two-dimensional results for the piston-driven model with the numerical predictions and the measurements with the pipe-mounted configuration, and presents the effect of geometry on the acoustic characteristics of dual Helmholtz resonators. The study is concluded with final remarks in Section 6.

## 2. Lumped-parameter theory

In light of the low frequencies of interest in the present study, the geometrical dimensions considered here are significantly smaller than the relatively long wavelengths. Hence, the spatial resolution is ignored next to develop expressions for both the resonance frequencies and transmission loss for a dual Helmholtz resonator installed in a side branch orientation as shown in Fig. 1.

### 2.1. Resonance frequencies of dual Helmholtz resonators

Applying the Newton's second law of motion to the first “mass-spring” of the lumped system, consisting of the first neck and cavity designated by subscript 1, gives

$$m_1 \frac{d^2 x_{C1}}{dt^2} = \Delta p_1 A_{C1} \equiv \frac{\Delta p_1}{\Delta \rho_1} \Delta \rho_1 A_{C1}, \quad (1)$$

where  $m_1 = \rho_o A_{C1} \ell'_{C1}$  is the mass of the fluid in the neck,  $A_{C1}$  and  $\ell'_{C1}$  are the cross-sectional area and the neck length with end correction [18],  $p$  and  $\rho$  are the pressure and density, respectively, with subscript “o” denoting the mean value. In the limit of small variations, the right-hand side of Eq. (1) approaches

$$\frac{dp_1}{d\rho_1} d\rho_1 A_{C1} = c_o^2 d\rho_1 A_{C1} = -c_o^2 \rho_o^2 A_{C1} dv_1, \quad (2)$$

where  $d\rho_1 = -\rho_o^2 dv_1$  is utilized for the last term,  $v_1 = V_1/M_1$  with  $V$  being the volume of the cavity and  $M$  the mass of the gas contained inside, and the speed of sound is designated by  $c_o$  ( $\approx 340.3$  m/s throughout the present study). Then, combining

$$dv_1 = \frac{dV_1}{M_1} = \frac{A_{C1} x_{C1} - A_{C2} x_{C2}}{M_1} \quad (3)$$

with Eqs. (1) and (2) and rearranging, in view of  $M_1 = \rho_o V_1$ , yield

$$\frac{d^2 x_{C1}}{dt^2} + \frac{c_o^2 A_{C1}}{\ell'_{C1} V_1} x_{C1} - \frac{c_o^2 A_{C2}}{\ell'_{C1} V_1} x_{C2} = 0. \quad (4)$$

Applying the Newton's second law to the second “mass-spring” of the lumped system, consisting of the second neck and cavity, designated by subscript 2, gives

$$m_2 \frac{d^2 x_{C2}}{dt^2} = \Delta p_2 A_{C2} - \Delta p_1 A_{C2} = \frac{\Delta p_2}{\Delta \rho_2} \Delta \rho_2 A_{C2} - \frac{\Delta p_1}{\Delta \rho_1} \Delta \rho_1 A_{C2}, \quad (5)$$

where  $m_2 = \rho_o A_{C2} \ell'_{C2}$  is the mass of the fluid in the neck. Again, in the limit of small variations, and noting  $d\rho_2 = -\rho_o^2 dv_2$ , Eq. (5) becomes

$$m_2 \frac{d^2 x_{C2}}{dt^2} = -c_o^2 \rho_o^2 A_{C2} dv_2 + c_o^2 \rho_o^2 A_{C2} dv_1. \quad (6)$$

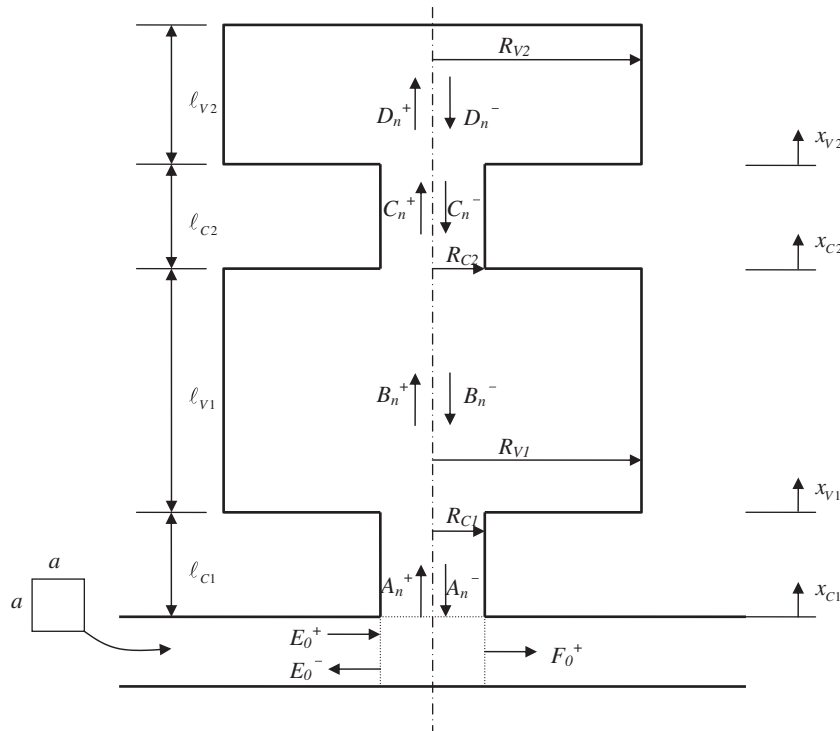


Fig. 1. Dual Helmholtz resonator.

Substituting  $dv_1$  from Eq. (3) and  $dv_2 = \frac{dv_2}{M_2} = \frac{A_{C2}x_{C2}}{M_2}$  into Eq. (6) with  $M_2 = \rho_o V_2$  and rearranging yields

$$\frac{d^2 x_{C2}}{dt^2} + \frac{c_o^2 A_{C2}}{\ell'_{C2} V_2} x_{C2} - \frac{c_o^2 A_{C1}}{\ell'_{C2} V_1} x_{C1} + \frac{c_o^2 A_{C2}}{\ell'_{C2} V_1} x_{C2} = 0. \quad (7)$$

Inserting  $x_{C1} = X_{C1} e^{i\omega t}$  and  $x_{C2} = X_{C2} e^{i\omega t}$  into Eqs. (4) and (7), and combining them for the angular frequency  $\omega$  gives

$$\omega^4 - \left( \frac{A_{C1}}{\ell'_{C1} V_1} + \frac{A_{C2}}{\ell'_{C2} V_1} + \frac{A_{C2}}{\ell'_{C2} V_2} \right) c_o^2 \omega^2 + \frac{A_{C1}}{\ell'_{C1} V_1} \frac{A_{C2}}{\ell'_{C2} V_2} c_o^4 = 0. \quad (8)$$

The roots of Eq. (8) reveal the angular resonance frequencies,  $\omega = 2\pi f$ . The resonance frequencies for a dual Helmholtz resonator are then obtained as

$$f_{1,2} = \frac{c_o}{2\sqrt{2}\pi} \sqrt{\left( \frac{A_{C1}}{\ell'_{C1} V_1} + \frac{A_{C2}}{\ell'_{C2} V_1} + \frac{A_{C2}}{\ell'_{C2} V_2} \right) \pm \sqrt{\left( \frac{A_{C1}}{\ell'_{C1} V_1} + \frac{A_{C2}}{\ell'_{C2} V_1} + \frac{A_{C2}}{\ell'_{C2} V_2} \right)^2 - 4 \frac{A_{C1}}{\ell'_{C1} V_1} \frac{A_{C2}}{\ell'_{C2} V_2}}} \quad (9)$$

which may be rearranged, in terms of  $\alpha_1 = \frac{A_{C1}}{\ell'_{C1} V_1}$ ,  $\alpha_2 = \frac{A_{C2}}{\ell'_{C2} V_2}$ ,  $\alpha_{21} = \frac{\alpha_2}{\alpha_1}$  and  $V_{21} = \frac{V_2}{V_1}$  as

$$\frac{f_{1,2}}{\frac{c_o}{2\pi} \sqrt{\alpha_1}} = \frac{1}{\sqrt{2}} \sqrt{1 + \alpha_{21}(1 + V_{21}) \pm \sqrt{[1 + \alpha_{21}(1 + V_{21})]^2 - 4\alpha_{21}}}. \quad (10)$$

Note for example, as  $\alpha_2 \rightarrow 0$ , the resonance frequency defaults, as expected, to

$$f = \frac{c_o}{2\pi} \sqrt{\alpha_1}, \quad (11)$$

the well-known lumped limit of single Helmholtz resonator.

## 2.2. Transmission loss (TL) for dual Helmholtz resonators

Applying pressure  $p$  to the mass in the “first” neck, the forced vibration of the fluid in this neck–cavity is expressed as

$$m_1 \frac{d^2 x_{C1}}{dt^2} = p A_{C1} - \frac{\rho_o c_o^2 A_{C1}}{V_1} (A_{C1} x_{C1} - A_{C2} x_{C2}). \quad (12)$$

Similarly, for the mass in the “second” neck,

$$m_2 \frac{d^2 x_{C2}}{dt^2} = \frac{\rho_o c_o^2 A_{C2}}{V_1} (A_{C1} x_{C1} - A_{C2} x_{C2}) - \frac{\rho_o c_o^2 A_{C2}^2}{V_2} x_{C2}. \quad (13)$$

Substituting  $x_{C1} = X_{C1} e^{i\omega t}$ ,  $x_{C2} = X_{C2} e^{i\omega t}$ , and  $p = P e^{i\omega t}$  into Eqs. (12) and (13) and rearranging in the matrix form yields

$$\mathbf{K} \begin{Bmatrix} X_{C1} \\ X_{C2} \end{Bmatrix} = \begin{Bmatrix} P A_{C1} \\ 0 \end{Bmatrix}, \quad (14)$$

where

$$\mathbf{K} = \begin{bmatrix} k_{11} & k_{12} \\ k_{21} & k_{22} \end{bmatrix} = \begin{bmatrix} -m_1 \omega^2 + \frac{\rho_o c_o^2 A_{C1}^2}{V_1} & -\frac{\rho_o c_o^2 A_{C1} A_{C2}}{V_1} \\ -\frac{\rho_o c_o^2 A_{C1} A_{C2}}{V_1} & -m_2 \omega^2 + \rho_o c_o^2 A_{C2}^2 \left( \frac{1}{V_1} + \frac{1}{V_2} \right) \end{bmatrix}, \quad (15)$$

is the stiffness matrix. By manipulating Eq. (14),

$$\frac{X_{C1}}{P A_{C1}} = \frac{k_{22}}{k_{11} k_{22} - k_{12} k_{21}}, \quad \frac{X_{C2}}{P A_{C1}} = \frac{-k_{12}}{k_{11} k_{22} - k_{12} k_{21}}. \quad (16)$$

In view of  $u_{C1} = \frac{dx_{C1}}{dt} = i\omega x_{C1}$ , Eq. (16) gives the normalized specific impedance of the Helmholtz resonator as

$$Z_H = \frac{p}{\rho_o c_o u_{C1}} = \frac{1}{i\omega \rho_o c_o} \frac{P}{X_{C1}} = \frac{1}{i\omega \rho_o c_o A_{C1}} \frac{k_{11} k_{22} - k_{12} k_{21}}{k_{22}}, \quad (17)$$

which yields the transmission loss of a dual Helmholtz resonator, through Ref. [13], as

$$TL = 20 \log_{10} \left| 1 + \frac{A_{C1}}{2A_p} \frac{1}{Z_H} \right|, \quad (18)$$

with  $A_p$  being the cross-sectional area of the main pipe. From a detailed derivation in Appendix B, TL is then expressed as

$$TL = 20 \log_{10} \left| 1 + \frac{A_{C1}}{2A_p} \frac{1}{ik \ell'_{C1} + \frac{A_{C1}}{ik V_1} \left( 1 - \frac{A_{C2} V_2}{A_{C2} V_2 + A_{C2} V_1 - V_1 V_2 k^2 \ell'_{C2}} \right)} \right| \quad (19)$$

or, by rearranging,

$$TL = 20 \log_{10} \left| 1 + i \frac{A_{C1}}{2A_p} \left\{ k \ell'_{C1} \left[ -1 + \frac{\alpha_1}{k^2} \left( 1 - \frac{1}{1 + \frac{1}{V_{21}} - \frac{k^2}{\alpha_2} \frac{1}{V_{21}}} \right) \right] \right\}^{-1} \right|. \quad (20)$$

## 3. Two-dimensional analytical approach

### 3.1. Solution for circular neck and volume

The pressure and axial particle velocity for the two-dimensional sound propagation in the first neck (recall Fig. 1) can be written as

$$P_A(r, x) = \sum_{n=0}^{\infty} (A_n^+ e^{-ik_{x,A,n} x} + A_n^- e^{ik_{x,A,n} x}) \psi_{A,n}(r), \quad (21)$$

and

$$u_A(r, x) = \frac{1}{\rho_o \omega} \sum_{n=0}^{\infty} k_{x,A,n} (A_n^+ e^{-ik_{x,A,n} x} - A_n^- e^{ik_{x,A,n} x}) \psi_{A,n}(r), \quad (22)$$

where  $n$  denotes the radial mode number;  $A_n^+$  are  $A_n^-$  the modal amplitudes corresponding to waves traveling in the positive and negative  $x$ -directions;

$$\psi_{A,n}(r) = J_0(k_{r,A,n} r), \quad (23)$$

is the eigenfunction with  $k_{r,A,n}$  being the radial wavenumber satisfying the boundary condition of

$$J_0(k_{r,A,n} R_{C1}) = 0, \quad (24)$$

$R_{C1}$  being the radius of the first neck; and  $k_{x,A,n}$  the axial wavenumber related to the radial wavenumber by

$$k_{x,A,n}^2 + k_{r,A,n}^2 = k_o^2 = (\omega/c_o)^2. \quad (25)$$

The sound pressure and axial particle velocities for the remaining circular rigid ducts (first cavity, second neck, and second cavity with their respective coefficients  $B_n^{\pm}$ ,  $C_n^{\pm}$  and  $D_n^{\pm}$ ) can be expressed similar to Eqs. (21) and (22).

### 3.2. Boundary conditions and TL

The coefficients  $A_n^{\pm}$  through  $D_n^{\pm}$  can be determined by the boundary conditions. At  $x_{V2} = \ell_{V2}$ , the rigid boundary condition,  $u_D = 0$ , gives

$$D_n^+ = D_n^- e^{2ik_{D,n}\ell_{V2}}. \quad (26)$$

At  $x_{C2} = \ell_{C2}$  or  $x_{V2} = 0$ , the pressure and volume velocity continuity conditions between the second neck and cavity provide [12], respectively,

$$[C_s^+ e^{-ik_{Cs}\ell_{C2}} + C_s^- e^{ik_{Cs}\ell_{C2}}] \langle \psi_{C,s}, \psi_{C,s} \rangle_C - \sum_{n=0}^{\infty} [D_n^+ + D_n^-] \langle \psi_{D,n}, \psi_{C,s} \rangle_C = 0, \quad (27)$$

$$k_{C,s} [C_s^+ e^{-ik_{Cs}\ell_{C2}} - C_s^- e^{ik_{Cs}\ell_{C2}}] \langle \psi_{C,s}, \psi_{C,s} \rangle_C - \left( \frac{R_{V2}}{R_{C2}} \right)^2 \sum_{n=0}^{\infty} k_{D,n} [D_n^+ - D_n^-] \langle \psi_{D,n}, \psi_{C,s} \rangle_C = 0, \quad (28)$$

where  $s = 0, 1, 2, \dots, \infty$ , and  $\langle \rangle_s$  is the integration over area  $S$  which can be expressed as

$$\langle \psi_{C,s}, \psi_{C,s} \rangle_C = \frac{R_{C2}^2}{2} J_0^2(\alpha_s), \quad (29)$$

and

$$\langle \psi_{D,n}, \psi_{C,s} \rangle_C = \begin{cases} \frac{R_{C2} R_{V2} J_0(\alpha_s) J_1\left(\alpha_n \frac{R_{C2}}{R_{V2}}\right)}{\left(\frac{\alpha_n}{R_{V2}}\right)^2 - \left(\frac{\alpha_s}{R_{C2}}\right)^2} & \frac{\alpha_n}{R_{V2}} \neq \frac{\alpha_s}{R_{C2}}, \\ \frac{R_{C2}^2}{2} J_0^2(\alpha_s) & \frac{\alpha_n}{R_{V2}} = \frac{\alpha_s}{R_{C2}} \end{cases}, \quad (30)$$

where  $\alpha_{n \text{ or } s}$  are the roots satisfying the rigid wall boundary condition of

$$J_0'(\alpha_{n \text{ or } s}) = J_1(\alpha_{n \text{ or } s}) = 0. \quad (31)$$

Four additional equations can be obtained from the remaining boundary conditions between second neck and first volume ( $x_{C2} = 0$ ), and first volume and first neck ( $x_{V1} = 0$ ) which may be expressed in a form similar to Eqs. (27) and (28). Further details of these equations will be avoided here for the brevity. In view of planar wave propagation in the main duct, the boundary conditions (pressure and volume velocity continuity) at the junction between the main duct and first neck yield

$$E_0^+ + E_0^- = A_0^+ + A_0^- = F_0^+, \quad (32)$$

and

$$A_p(E_0^+ - E_0^-) = A_{C1}(A_0^+ - A_0^-) + A_p F_0^+. \quad (33)$$

Ten equations [Eqs. (26)–(28), (32) and (33), and four additional ones mentioned earlier] define the relationships among eleven coefficients ( $A_n^\pm - D_n^\pm$ ,  $E_0^\pm$ , and  $F_0^\pm$ ), finally leading to the determination of transmission loss by

$$TL = 20 \log_{10} \left| \frac{E_0^+}{F_0^+} \right|. \quad (34)$$

#### 4. Boundary element method

The boundary element method allows accurate predictions of the acoustic performance of silencers with complex geometry and/or in the presence of multi-dimensional wave propagation [12,19]. The wave propagation is governed by the Helmholtz equation

$$\nabla^2 p(x, y, z) + k^2 p(x, y, z) = 0, \quad (35)$$

where the integral form for an acoustic domain is defined using the Green's theorem as

$$C(q_1)P(q_1) = \int_{\Gamma} \left[ G(q_1, q_2) \frac{\partial P(q_2)}{\partial \vec{n}} - P(q_2) \frac{\partial G(q_1, q_2)}{\partial \vec{n}} \right] d\Gamma(q_2), \quad (36)$$

where  $q_1$  and  $q_2$  are points on the boundary surface  $\Gamma$ ,  $C(q_1)$  denotes coefficients, and

$$G(q_1, q_2) = \frac{e^{-ik|q_1 - q_2|}}{4\pi|q_1 - q_2|} \quad (37)$$

is the Green's function (or fundamental solution). By solving Eq. (35) numerically and imposing zero velocity on the rigid surfaces, the transfer matrix, which expresses the relationship between acoustic pressure and particle velocity at the inlet and outlet of the main duct, may be determined by

$$\begin{bmatrix} p_{in} \\ \rho_0 c_0 u_{in} \end{bmatrix} = \begin{bmatrix} T_{11} & T_{12} \\ T_{21} & T_{22} \end{bmatrix} \begin{bmatrix} p_{out} \\ \rho_0 c_0 u_{out} \end{bmatrix}, \quad (38)$$

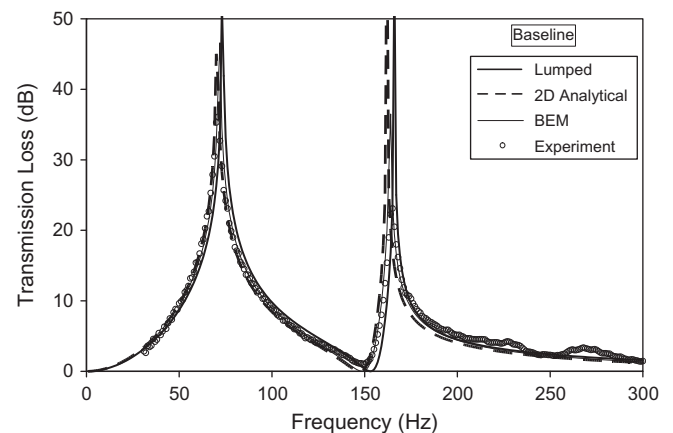
where  $T_{ij}$  represents transfer matrix elements. Transmission loss may then be calculated from this transfer matrix by

$$TL = 20 \log_{10} \left( \frac{1}{2} |T_{11} + T_{12} + T_{21} + T_{22}| \right). \quad (39)$$

#### 5. Results and discussion

For an experimental comparison, a base prototype (hereafter referred to as simply “baseline”) has been fabricated with  $R_{C1} = 2$  cm,  $R_{V1} = 7.62$  cm,  $R_{C2} = 1.75$  cm,  $R_{V2} = 7.62$  cm,  $\ell_{C1} = 8.5$  cm,  $\ell_{V1} = 20.32$  cm,  $\ell_{C2} = 7.62$  cm and  $\ell_{V2} = 10.16$  cm with the corresponding volume of the first and second cavities being  $V_{1b} = 3706$  cm<sup>3</sup> and  $V_{2b} = 1853$  cm<sup>3</sup> respectively. The dimensions of the 1st neck–cavity and the main duct match those of the resonator in Refs. [12,13]. The main duct is built of a square cross-section (4.3 cm × 4.3 cm) for a clear identification of the geometric neck length  $\ell_{C1}$ . The square main duct is then connected to a circular impedance tube with smooth transitions that retain a constant cross-sectional area development. The experimental apparatus to measure the transmission loss is based on two-microphone technique which employs two pairs of microphones (one pair before and another after the resonator) on an impedance-tube setup [20]. B&K 3550 multichannel analysis system has been used along with the random sound input. Throughout the frequency range of interest, the reflection coefficient measured on the downstream side of the resonator is ensured, by an appropriate termination, to remain below 0.1 (or, often significantly below this value) which translates to accurately measured transmission loss.

Fig. 2 compares the transmission loss from the lumped and 2D analytical approaches, BEM, and experiments with a pipe-mounted “baseline” configuration. Two TL peaks are observed at the resonance frequencies (73 and 166 Hz), as predicted by Eq. (9) [or,



**Fig. 2.** Transmission loss of the “baseline” dual Helmholtz resonator from lumped-parameter theory, 2D analytical predictions, BEM, and measurements ( $R_{C1} = 2$  cm,  $R_{V1} = 7.62$  cm,  $R_{C2} = 1.75$  cm,  $R_{V2} = 7.62$  cm,  $\ell_{C1} = 8.5$  cm,  $\ell_{V1} = 20.32$  cm,  $\ell_{C2} = 7.62$  cm, and  $\ell_{V2} = 10.16$  cm).

equivalently, Eq. (10)]. There is also a frequency in between where TL is nearly zero, suggesting an anti-resonance behavior. The predictions from the lumped approach show, in general, a good agreement with the experimental results in terms of frequency-dependency of TL. Such an agreement points to the appropriateness of the end corrections employed in this study. Deviations in the magnitude of TL between the lumped approach and experiments may be observed in the immediate vicinity of resonance frequencies. Such deviations may be attributed to the neglect of viscous damping in the Helmholtz resonator as well as the finite input power available to acoustic drivers in the experimental set-up. The results from the lumped model match the 2D analytical predictions reasonably well. Slightly lower resonance frequency predictions of the latter may be attributed to the fact that the end correction on the main duct side has not been accounted for by the nature of the piston-driven model. BEM captures the frequency-dependency of experimental TL very well, as expected (again with the exception of precise amplitudes right at the resonance frequencies for similar reasons). It should be noted that the intended target of the resonator examined in the present study is strictly the low frequencies which present a major challenge in terms of noise control in the breathing system of reciprocating engines and compressors. The effect of dual Helmholtz resonator is, therefore, shown here only over the low-frequency range (below 300 Hz), while exploring the acoustic performance of this configuration is readily (and accurately) doable at higher frequencies with multi-dimensional techniques such as BEM.

To investigate the impact of second neck–cavity, Fig. 3 presents TL from the lumped approach for four different resonators: (1) baseline, (2) 1st-neck–1st-cavity alone (“Sub-HR1”), (3) 2nd-neck–2nd-cavity alone (“Sub-HR2”), and (4) the one with 1st neck combined with a cavity having a volume as the sum of the 1st and 2nd cavities (“Sub-HR3”). Compared to the other three, the baseline configuration yields the lowest and highest resonance frequencies. The resonance frequency of Sub-HR1 is closer to the lower resonance frequency of the baseline, while that of Sub-HR2 to the higher resonance frequency. Hence, combining two resonators has moved the smaller of the two individual Sub-HR resonance frequencies to a lower value, and the larger of two to a higher value. The transmission loss of baseline around the first peak frequency is somewhat close to that of Sub-HR3.

To examine the effect of coupling between the two neck–cavities, Fig. 4 compares TL from the lumped approach of the baseline with another configuration (Hereafter referred to as “Alternative”), which has the same geometry as the former, except  $\ell_{C1} = 2.3$  cm

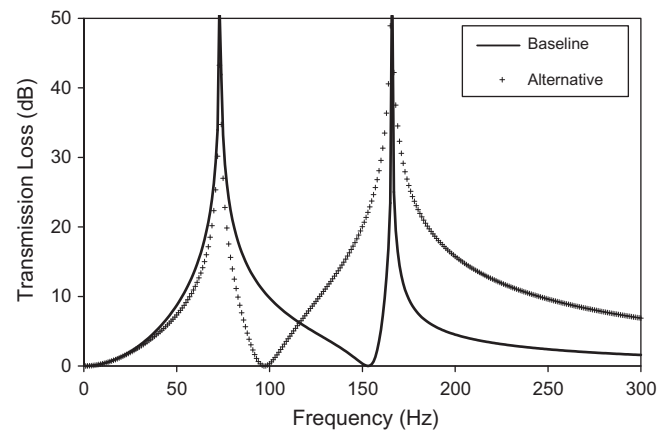


Fig. 4. Transmission loss prediction from lumped-parameter theory for baseline and alternative dual Helmholtz resonators.

and  $\ell_{C2} = 22$  cm. These neck lengths are chosen to retain the same resonance frequencies. The comparison in Fig. 4 indeed shows the same peak frequencies (73 and 166 Hz), while revealing different shapes near these two peaks for the two dual Helmholtz resonators. The modification of neck lengths is such that the resonance frequency of 2nd-neck–2nd-cavity alone of the alternative configuration is now lower than that of its counterpart with baseline, influencing in turn the TL behavior of the alternative. TL of baseline has a wider first peak (near 73 Hz), and narrower second peak (near 166 Hz) than alternative. For the baseline, TL is above 10 dB from 53 to 99 Hz (frequency width of 46 Hz), and 163 to 174 Hz (width 11 Hz); while for the alternative, the frequencies are from 58 to 82 Hz (frequency width of 24 Hz), and 127 to 245 Hz (width 118 Hz). The peak-width differences between the two dual Helmholtz resonators are due to the TL of the Sub-HRs. Next, a number of geometrical variations have been contrasted with the baseline, including cavities ( $\ell_{V1} = 22$  cm and  $\ell_{V2} = 9.4$  cm, and  $\ell_{V1} = 19.5$  cm and  $\ell_{V2} = 10.7$  cm); the first neck ( $\ell_{C1} = 6.25$  cm and  $R_{C1} = 1.75$  cm, and  $\ell_{C1} = 14$  cm and  $R_{C1} = 2.5$  cm); and the second neck ( $\ell_{C2} = 10.5$  cm and  $R_{C2} = 2$  cm, and  $\ell_{C2} = 5.25$  cm and  $R_{C2} = 1.5$  cm). The TL behavior of these different dual Helmholtz resonators are nearly identical, suggesting different geometric possibilities for the same attenuation characteristics.

The lumped and 2D analytical results as a function of the 2nd neck length [ $\ell_{C2} = 2.54, 5.08, 7.62$  (baseline), 10.16, and 12.70 cm]

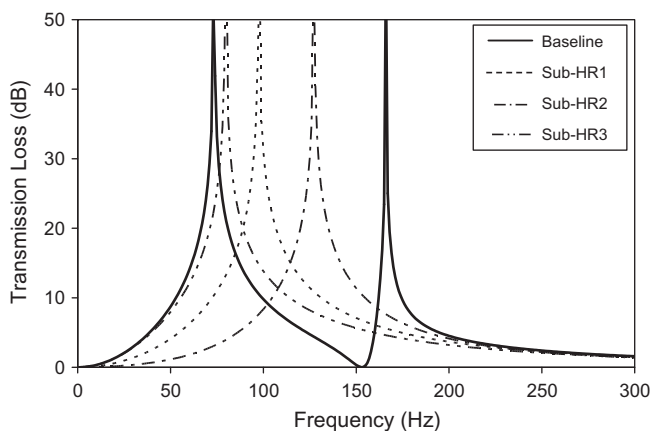


Fig. 3. Transmission loss prediction from lumped-parameter theory for four different resonators: Baseline, 1st-neck–1st-cavity (Sub-HR1), 2nd-neck–2nd-cavity (Sub-HR2), and one with the 1st neck and sum of the 1st and 2nd cavities (Sub-HR3).

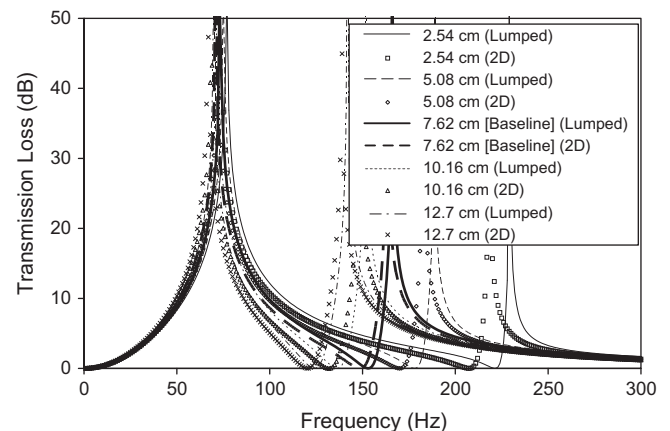


Fig. 5. Transmission loss prediction from the lumped and 2D analytical approach with varying length of 2nd neck [ $\ell_{C2} = 2.54, 5.08, 7.62$  (baseline), 10.16, and 12.70 cm].

are presented in Fig. 5. The other parameters are kept the same as the baseline. The second peak moves to lower frequencies with increasing  $\ell_{C2}$ , illustrating again the dominance of 2nd cavity on the higher resonance frequency for the baseline. The location of the first peak, however, is nearly constant (changes only from 77 Hz to 70 Hz). For all five resonators, the predictions from the lumped approach show a reasonable agreement with the 2D analytical results. The effect of the 2nd-neck radius [ $R_{C2} = 1.00, 1.25, 1.50, 1.75$  (baseline), and  $2.00$  cm] is illustrated in Fig. 6. The second peak moves to higher frequency with increasing  $R_{C2}$ , again illustrating the impact of 2nd cavity of baseline on the higher resonance frequency, while the movement in the first peak remains relatively limited (from 61 Hz to 75 Hz). For all configurations, the predictions from the lumped approach show a reasonable agreement with the 2D analytical results.

The results from the lumped approach with varying 2nd-cavity volume normalized with respect to  $V_{2b}$ ,  $V_2/V_{2b} = 0.5, 0.75, 1$  (baseline),  $1.25$ , and  $1.5$  are presented in Fig. 7. Both resonance frequencies are lowered with increasing  $V_2$ , and the first peak tends to move less than the second one. The effect of the 2nd-cavity shape [keeping the volume  $V_{2b}$  the same and changing the ratio of diameter  $D_{V2} = 2R_{V2}$  to length of the cavity to  $D_{V2}/\ell_{V2} = 0.25, 0.5, 1, 1.5$  (baseline), and  $2$ ] is studied using the 2D analytical approach. The results depicted in Fig. 8 show that this variation moves the second resonance frequency by a small amount (about 7 Hz),

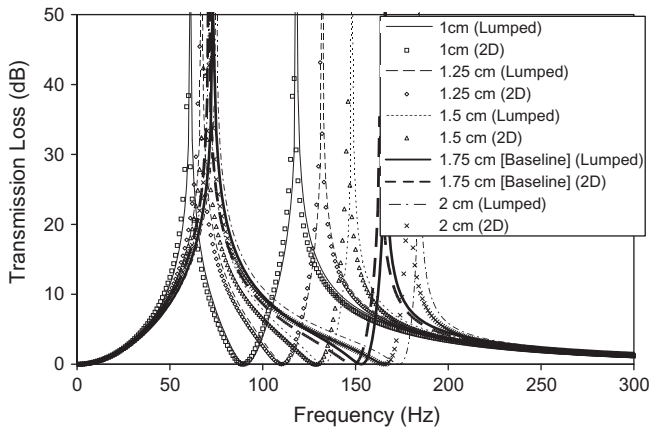


Fig. 6. Transmission loss prediction from the lumped and 2D analytical approach with varying radius of the 2nd neck [ $R_{C2} = 1, 1.25, 1.50, 1.75$  (baseline), and  $2$  cm].

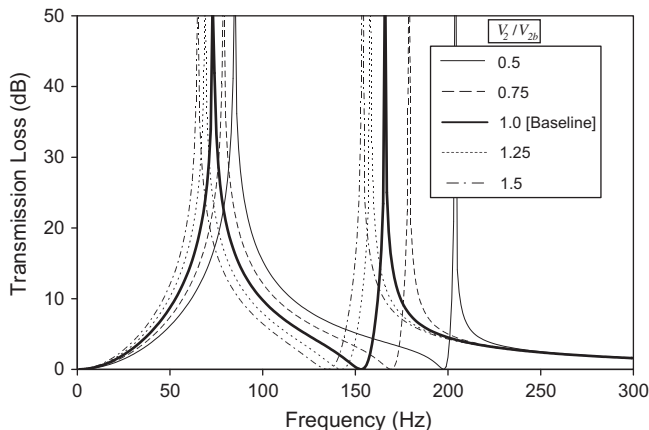


Fig. 7. Transmission loss prediction from the lumped-parameter theory with varying volume of 2nd cavity:  $V_2/V_{2b} = 0.5, 0.75, 1$  (baseline),  $1.25$ , and  $1.5$ .

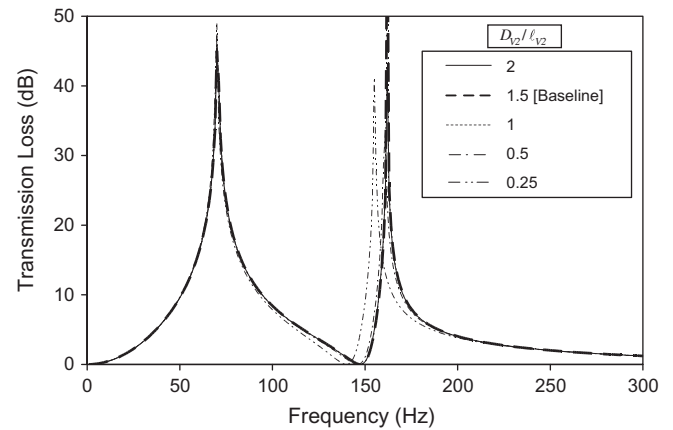


Fig. 8. Transmission loss prediction from 2D analytical approach with varying shape (different ratio of  $D_{V2}/\ell_{V2}$ ) of 2nd cavity.

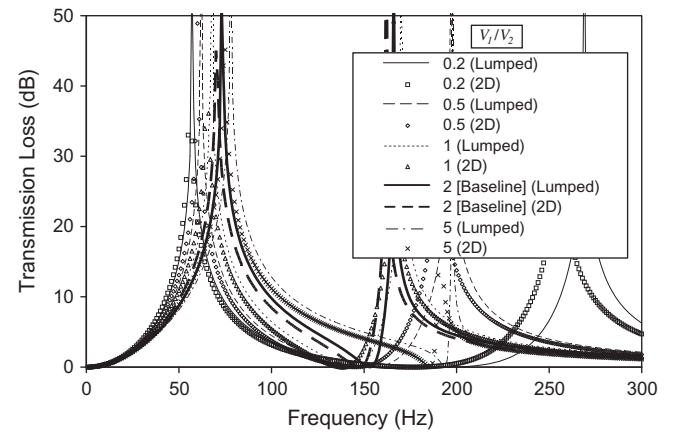


Fig. 9. Transmission loss prediction from the lumped and 2D analytical approach with varying volume ratio of the two cavities:  $V_1/V_2 = 0.2, 0.5, 1, 2$ , and  $5$ .

whereas the first peak essentially remains the same. This is expected in view of the fact that frequencies considered in this study are too low to promote strong multi-dimensional wave propagation. The effect of the volume ratio of the two cavities (while maintaining  $R_{V1}$ ,  $R_{V2}$  and the total cavity volume  $V_{1b} + V_{2b} = 5559 \text{ cm}^3$ )  $V_1/V_2 = 0.2, 0.5, 1, 2$ , and  $5$  is illustrated in Fig. 9. The first resonance frequency is lowered with the decreasing volume ratio, while the second resonance frequency exhibits a more significant shift in the opposite direction.

## 6. Concluding remarks

The acoustic performance of a dual Helmholtz resonator composed of two Helmholtz resonators in series (neck-cavity-neck-cavity) is investigated. Closed-form relationships have been developed for the resonance frequencies and the transmission loss of this two degree-of-freedom acoustic configuration by using a lumped approach. Comparisons of the transmission loss and resonance frequencies from the lumped and 2D analytical approaches, a numerical method (BEM), and experiments show a reasonable agreement. The effects of the 2nd neck and cavity have been illustrated: the resonance frequencies decrease with the increasing length of the 2nd neck, decreasing radius of the 2nd neck, and increasing volume of the 2nd cavity; while the influence of the shape of the 2nd cavity (diameter to length ratio) remains small.

The first resonance frequency may also be lowered with decreasing  $V_1/V_2$  while retaining the total volume of the two cavities. The variation in the first resonance frequency is observed to be smaller than the second one. It has also been demonstrated, through variation of neck lengths, the width of TL can be adjusted, while keeping the peak frequencies the same. Nearly identical TL can be obtained for dual Helmholtz resonators of same dimensions, but with different (a) radii of the cavities; (b) length and radius of the first neck; or (c) length and radius of the second neck, suggesting multiple solutions to achieve similar acoustical performance. Such flexibility could prove to be helpful particularly when dealing with space constraints in a variety of engineering applications.

## Appendix A. End correction

For a pipe of radius  $R_{\text{pipe}}$  ending with an infinite flange, the end correction  $\delta$  can be expressed as [10]

$$\frac{\pi}{4} < \frac{\delta}{R_{\text{pipe}}} < \frac{8}{3\pi}. \quad (\text{A.1})$$

An end correction that accounts for the higher order wave propagation effects between the neck and volume can be approximated by

$$\delta = 0.85R_{\text{neck}} \left( 1 - 1.25 \frac{R_{\text{neck}}}{R_{\text{volume}}} \right) \quad (\text{A.2})$$

which reduces to

$$\delta = 0.85R_{\text{neck}} \quad (\text{A.3})$$

as  $R_{\text{neck}}/R_{\text{volume}}$  becomes negligible.

To account for the higher order wave propagation effects between the circular neck and main pipe (one direction being infinite, while the size of the other direction is close to that of the neck), the end correction is approximated by

$$\delta = 0.82 \frac{R_{\text{neck}}}{2}, \quad (\text{A.4})$$

as  $R_{\text{neck}}/R_{\text{volume}}$  becomes negligible.

In this study, the end correction between the neck and volume is modeled by Eq. (A.2), and the end correction between the neck and the main duct by Eq. (A.4).

## Appendix B. Transmission loss

Substituting  $k$  values from Eq. (15) into Eq. (17) gives, for the specific acoustic impedance

$$Z_H = \frac{m_1 m_2 \omega^4 - \rho_0 c_0^2 \left[ m_1 A_{C2}^2 \left( \frac{1}{V_1} + \frac{1}{V_2} \right) + m_2 A_{C1}^2 \frac{1}{V_1} \right] \omega^2 + \frac{\rho_0^2 c_0^4 A_{C1}^2 A_{C2}^2}{V_1 V_2}}{i \omega \rho_0 c_0 A_{C1} \left[ -m_2 \omega^2 + \rho_0 c_0^2 A_{C2}^2 \left( \frac{1}{V_1} + \frac{1}{V_2} \right) \right]}. \quad (\text{B.1})$$

Inserting  $m_1 = \rho_0 A_{C1} \ell_{C1}$  and  $m_2 = \rho_0 A_{C2} \ell_{C2}$  into Eq. (B.1) and rearranging give

$$Z_H = \frac{\ell_{C1} \ell_{C2} \omega^4 - c_0^2 \omega^2 \left[ \ell_{C1} A_{C2} \left( \frac{1}{V_1} + \frac{1}{V_2} \right) + \ell_{C2} A_{C1} \frac{1}{V_1} \right] + \frac{c_0^4 A_{C1} A_{C2}}{V_1 V_2}}{i \omega c_0 \left[ -\ell_{C2} \omega^2 + c_0^2 A_{C2} \left( \frac{1}{V_1} + \frac{1}{V_2} \right) \right]} \quad (\text{B.2})$$

or, in terms of wavenumber  $k = \omega/c_0$ ,

$$Z_H = \frac{\ell_{C1} \ell_{C2} k^4 - k^2 \left[ \ell_{C1} A_{C2} \left( \frac{1}{V_1} + \frac{1}{V_2} \right) + \ell_{C2} A_{C1} \frac{1}{V_1} \right] + \frac{A_{C1} A_{C2}}{V_1 V_2}}{ik \left[ -\ell_{C2} k^2 + A_{C2} \left( \frac{1}{V_1} + \frac{1}{V_2} \right) \right]} = \frac{-k^2 \ell_{C1} \left[ -\ell_{C2} k^2 + A_{C2} \left( \frac{1}{V_1} + \frac{1}{V_2} \right) \right] - k^2 \ell_{C2} A_{C1} \frac{1}{V_1} + \frac{A_{C1} A_{C2}}{V_1 V_2}}{ik \left[ -\ell_{C2} k^2 + A_{C2} \left( \frac{1}{V_1} + \frac{1}{V_2} \right) \right]}. \quad (\text{B.3})$$

Further manipulation yields

$$\begin{aligned} Z_H &= ik \ell_{C1} + \frac{\frac{A_{C1} A_{C2}}{V_1 V_2} - k^2 \ell_{C2} \frac{A_{C1}}{V_1}}{ik \left[ -\ell_{C2} k^2 + A_{C2} \left( \frac{1}{V_1} + \frac{1}{V_2} \right) \right]} \\ &= ik \ell_{C1} + \frac{A_{C1}}{ik V_1} \frac{\frac{A_{C2}}{V_2} - k^2 \ell_{C2}}{\frac{A_{C2}}{V_1} + \frac{A_{C2}}{V_2} - k^2 \ell_{C2}} \\ &= ik \ell_{C1} + \frac{A_{C1}}{ik V_1} \left[ 1 - \frac{A_{C2} V_2}{A_{C2} V_2 + A_{C2} V_1 - V_1 V_2 k^2 \ell_{C2}} \right]. \end{aligned} \quad (\text{B.4})$$

Finally, substituting Eq. (B.4) into Eq. (18) gives the transmission loss as

$$TL = 20 \log_{10} \left| 1 + \frac{A_{C1}}{2A_p} \frac{1}{ik \ell_{C1} + \frac{A_{C1}}{ik V_1} \left[ 1 - \frac{A_{C2} V_2}{A_{C2} V_2 + A_{C2} V_1 - V_1 V_2 k^2 \ell_{C2}} \right]} \right|. \quad (\text{B.5})$$

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